MateriaLens Cost Model

Illustrative LLM and retrieval cost assumptions for a proposed MVP

1. Purpose and status

This model estimates variable cost for a text-only MateriaLens query. It is a planning model, not a measured bill or a complete hosting budget. All token volumes and usage rates are assumptions until tested with representative queries. Model pricing should be rechecked against the provider's current official pricing before external use.

2. Cost formula

LLM cost per query = (input tokens ÷ 1,000,000 × input price per 1M tokens) + (output tokens ÷ 1,000,000 × output price per 1M tokens).

Total variable cost per query = LLM cost + embedding/query cost + allocated vector/keyword retrieval and reranking cost.

Monthly variable cost per user = total variable cost per query × queries per user per month.

3. Working assumptions

Input	Planning assumption	Status / note
LLM	Gemini 2.5 Flash	Proposed model from the PRD; verify exact model ID and current pricing before calculation.
Input tokens per query	4,000 tokens	Illustrative allowance for system instructions, user query and retrieved evidence.
Output tokens per query	600 tokens	Illustrative concise answer plus Evidence Matrix.
Queries per user per month	330	Illustrative usage assumption; approximately 11 queries per day.
Input price	$0.30 per 1M tokens	Illustrative standard paid-tier text-input rate; confirm official pricing and applicable conditions.
Output price	$2.50 per 1M tokens	Illustrative standard paid-tier text-output rate; confirm official pricing and applicable conditions.
Embedding cost	Excluded from base arithmetic	Depends on chosen embedding model, whether documents are embedded once or repeatedly, and query volume.
Infrastructure	Excluded from base arithmetic	Hosting, database/vector store, logging, storage, monitoring, network and any reranking are not included.

4. Illustrative calculation

Component	Calculation	Estimated cost
Input	4,000 ÷ 1,000,000 × $0.30	$0.00120
Output	600 ÷ 1,000,000 × $2.50	$0.00150
LLM total per query	$0.00120 + $0.00150	$0.00270
LLM cost per user / month	330 × $0.00270	$0.891

Under these assumptions, LLM usage is approximately $0.0027 per query and $0.89 per user per month at 330 queries. This is not the full cost to serve a user. Actual cost can change with prompt size, retrieved passage count, output length, model pricing, caching and infrastructure.

5. Sensitivity to usage

Queries per user / month	LLM cost at $0.0027/query	Interpretation
100	$0.27	Light usage
330	$0.89	Base planning case
600	$1.62	Higher recurring usage
1,000	$2.70	Heavy usage; review latency and abuse controls as well as cost

6. Cost drivers not included

- One-time or repeated document embeddings during corpus ingestion and re-indexing.
- Vector database and keyword index hosting, storage, backups and network transfer.
- Optional reranker inference, OCR processing and document extraction.
- Application hosting, authentication, logs, monitoring, analytics and support.
- Retries, failed requests, long context windows, repeated prompts and provider-specific minimums or tier rules.
- Taxes, currency conversion and regional pricing differences.

7. Recommended optimization lever

Retrieve broadly enough to find relevant candidates, then pass only the strongest small set of passages to the LLM. This directly reduces input tokens while preserving the product's core requirement: source-backed answers. The trade-off is that overly aggressive filtering can remove useful context, so evaluate cost alongside Recall@5, citation support and groundedness rather than optimizing tokens alone.

8. Cost monitoring plan

- Log input and output tokens, model ID, latency and cost estimate per request.
- Report median and 95th-percentile tokens/cost per query, not only averages.
- Track cost by query type and identify unusually long prompts or retrieval contexts.
- Use caching only where source version and query semantics make reuse safe; invalidate cached answers when indexed content changes.
- Set usage limits and rate limits appropriate to the pilot; alert on unexpected increases.
- Recalculate this model whenever model pricing, prompt structure, Top-K or monthly usage assumptions change.

9. Verification checklist

☐ Confirm current official input/output rates for the exact model ID and account tier.

☐ Measure average input/output tokens on at least 30 representative queries.

☐ Estimate embedding and indexing costs from the actual corpus size.

☐ Estimate hosting and storage separately and disclose them in the total cost view.

☐ Re-run quality evaluations after any cost optimization that changes retrieval or context size.

10. Pricing reference

Pricing source to verify before submission: Google AI for Developers / Gemini API pricing — https://ai.google.dev/gemini-api/docs/pricing. Access date should be recorded when the rate is checked. The rates in this document are planning inputs and must not be treated as guaranteed current prices.